Homework 4, due Sep 17 before class

1. A metal sphere of radius r has dipolar magnetic field, Fig. 1. Imagine a cylindrical surface of radius $R > r$ (R is measured from the center of the sphere) aligned with the dipole. Some magnetic field lines close within the cylinder, some intersect the cylinder. Find polar angle θ that the last field line that closes inside the cylinder makes on the surface of the sphere.

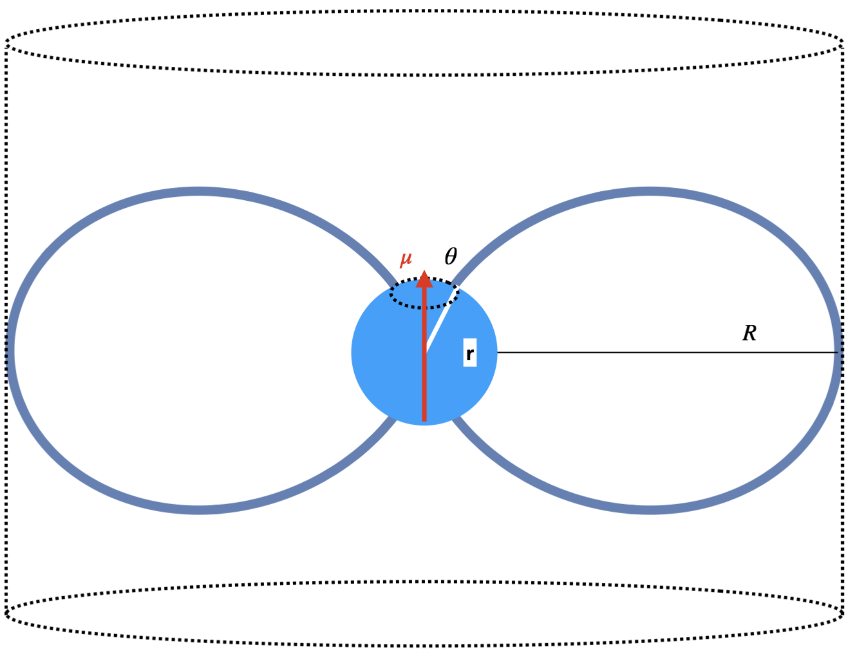

Fig. 1.— Dipole within a cylinder.

2. A metal sphere of radius R carries surface current $g_\phi = g_0 \sin\theta$. Find magnetic field inside and outside. Hint: look for $A_\phi \propto f(r)\sin\theta$, find solutions of $\Delta \mathbf{A} = 0$ inside and outside, then radial component of the magnetic field should be continuous, while the tangential component experiences a jump

$$B_\theta^{(+)} - B_\theta^{(-)} = \frac{4\pi}{c} g \tag{1}$$